

Stabilization of a Simulated Triple Inverted Pendulum using Reinforcement Learning

Introduction

The inverted pendulum on a cart is a classic benchmark problem in control theory due to its highly non-linear and unstable nature. While the single and double variants have been extensively researched, the triple inverted pendulum increases the complexity of the state space and thus makes stabilization much harder. This project aims to replace a feedback control loop by designing and training a reinforcement learning agent to autonomously discover and implement control policies capable of achieving stabilization, demonstrating the effectiveness of modern continuous-control reinforcement learning algorithms on highly dynamic systems.

Problem Formulation

The environment is comprised of a cart constrained to one-dimensional horizontal motion, with three interconnected pendulum links attached to it, constrained to planar motion. This system is then simulated using a physics engine (MuJoCo or PyBullet). Important mathematical objects are:

- **State Space:** A vector representing the system's kinematics. This includes the cart's position and velocity, as well as the angle and angular velocity of each of the three pendulum links.
- **Action Space:** A one dimensional vector representing the horizontal force applied on the cart.

- **Reward Function:** A function engineered to prioritize keeping the pendulum links upright, while penalizing excessive force and large deviations from the center.

Methodology

Because the action space (force) and state space are continuous, the project will utilize state-of-the-art Actor-Critic methods designed for continuous control.

- **Primary Algorithm:** Soft Actor-Critic (SAC) or Proximal Policy Optimization (PPO). SAC is favored for its sample efficiency and entropy-regularized exploration, which helps prevent converging to sub-optimal local minima in complex physical tasks.
- **Training Strategy:** The primary objective focuses on stabilization. The agent will be initialized randomly in a near-equilibrium state (e.g., all links within $\pm 5^\circ$ of the vertical). Episodes terminate when the cart exceeds track bounds or when any pendulum angle falls beyond a non-recoverable threshold (e.g., $\pm 15^\circ$ from the vertical).

Evaluation

The success of the trained policy will be evaluated based on:

- **Mean Episode Return:** The average cumulative reward over a set of evaluation episodes.
- **Survival Time:** The average time the agent can successfully keep the system upright before failure.
- **Robustness:** The agent's ability to recover from simulated disturbances (e.g., applying instantaneous random force impulses to the pendulums during evaluation).

Future Work

While the primary objective is the stabilization of the fully upright configuration from a near-equilibrium state, the highly nonlinear dynamics of the triple pendulum present several avenues for advanced RL research beyond the scope of the initial project:

- **Full Swing-Up Maneuver:** Transitioning the system from a stable, downward-hanging resting state (180°) to the fully upright unstable equilibrium. Due to the chaotic nature of the triple pendulum, this introduces a severe exploration challenge.
- **Alternative Equilibrium Configurations:** The fully upright ("up-up-up") position is only one of several unstable equilibria. Other configurations exist, such as balancing the first link upright, the second link hanging down, and the third link upright (an "up-down-up" configuration).